

ECUACIONES DIFERENCIALES II

Problemas 7

(34) En cada caso justifica la existencia y diferenciabilidad de $x(t, \lambda)$ en un entorno de $\mathbb{R} \times \{0\}$ y calcula $\frac{\partial x}{\partial \lambda}(t, 0)$.

(i) $\ddot{x} + x^3 = 0$, $x(0) = \lambda$, $\dot{x}(0) = 0$

(ii) $\ddot{x} + x^3 = 0$, $x(0) = 0$, $\dot{x}(0) = \lambda$

(iii) $\ddot{x} + \lambda x^3 = 0$, $x(0) = \dot{x}(0) = 0$

(iv) $\ddot{x} + x^3 = \lambda$, $x(0) = \dot{x}(0) = 0$.

(35) Se considera la ecuación de un péndulo con fricción

$$\ddot{\theta} + c \dot{\theta} + \frac{g}{l} \sin \theta = 0$$

y se supone que el coeficiente $c > 0$ cumple la condición de fricción débil

$$c^2 < \frac{4g}{l}.$$

Sea $\theta(t, \varepsilon)$ la solución que cumple las condiciones iniciales

$$\theta(0) = 0, \quad \dot{\theta}(0) = \varepsilon.$$

Demuestra

(i) $\theta(t, \varepsilon)$ es única y está bien definida en $]-\infty, +\infty[$

(ii) $(t, \varepsilon) \in \mathbb{R} \times \mathbb{R} \mapsto \theta(t, \varepsilon) \in \mathbb{R}$ es diferenciable

(iii) $\theta(t, \varepsilon) = \frac{\varepsilon}{\omega} e^{-\frac{c}{2}t} \sin \omega t + R(t, \varepsilon)$

con $\omega = \frac{1}{2} \sqrt{\frac{4g}{l} - c^2}$ y $\lim_{\varepsilon \rightarrow 0} \frac{R(t, \varepsilon)}{\varepsilon} = 0$

para cada $t \in \mathbb{R}$.

(36) Sea $x(t; a, b)$ la solución del problema

$$\dot{x} + \sin(ax) = t, \quad x(0) = b.$$

Demuestra

(i) $x(t; a, b)$ es única y está definida en $]-\infty, +\infty[$

(ii) $(t; a, b) \in \mathbb{R}^3 \mapsto x(t; a, b) \in \mathbb{R}$ es diferenciable

(iii) Calcula $\frac{\partial x}{\partial a}(t; 0, 1)$ y $\frac{\partial x}{\partial b}(t; 0, 1)$.

(37) Se considera la función $F(y) = \int_0^1 e^{-x^2} \cos 2xy \, dx$.

Encuentra una ecuación diferencial lineal de la que $F(y)$ es solución.

(34) En cada caso justifica la existencia y diferenciabilidad de $x(t, \lambda)$ en un entorno de $\mathbb{R} \times \{0\}$ y calcula $\frac{\partial x}{\partial \lambda}(t, 0)$.

(i) $\ddot{x} + x^3 = 0, \quad x(0) = \lambda, \quad \dot{x}(0) = 0$

Sea $x_1 = x, x_2 = \dot{x}, z = (x_1, x_2)$

$\dot{x}_1 = \dot{x} = x_2, \quad \Sigma: D = \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R} / \Sigma(t, z) = \begin{pmatrix} x_2 \\ -x_1^3 \end{pmatrix} \Rightarrow \Sigma \in C^1(D) \Rightarrow$ hay unicidad.

$\dot{x}_2 = \dot{\dot{x}} = -x_1^3$

$\frac{\partial \Sigma}{\partial z}(t, z) = \begin{pmatrix} 0 & 1 \\ -3x_1^2 & 0 \end{pmatrix} \text{ cont.} \Rightarrow z \text{ es diag.} \quad \frac{\partial \Sigma}{\partial z}(t, z(t, 0)) = \begin{pmatrix} 0 & 1 \\ -3x_1(t, 0)^2 & 0 \end{pmatrix}$

Consideramos $\begin{cases} \ddot{x} + x^3 = 0 \\ x(0) = \dot{x}(0) = 0 \end{cases}$ vemos $x(t, 0) = 0$ es sol. Sea $V(t) = \begin{pmatrix} \frac{\partial x_1}{\partial \lambda}(t, 0) \\ \frac{\partial x_2}{\partial \lambda}(t, 0) \end{pmatrix}$

Por tanto, $\frac{\partial \Sigma}{\partial z}(t, z(t, 0)) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = A$

$\frac{dz}{dt}(t, 0) = \frac{\partial z}{\partial t}(t, 0) \cdot \frac{d}{d\lambda}(0) + \frac{\partial z}{\partial \lambda}(t, 0) \cdot \frac{d\lambda}{d\lambda} = V(t) \Rightarrow V(0) = \frac{dz}{d\lambda}(0, 0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\dot{V}(t) = A V(t) \quad \left. \begin{matrix} V(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{matrix} \right\} \Rightarrow \begin{pmatrix} \dot{V}_1 \\ \dot{V}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} \Rightarrow \begin{matrix} \dot{V}_1 = V_2 \\ \dot{V}_2 = 0 \end{matrix} \quad \left. \begin{matrix} \end{matrix} \right\} \Rightarrow$

$V_2(t) = C_1, V_1(t) = C_1 t + C_2 \Rightarrow \begin{cases} C_2 = 1 \\ C_1 = 0 \end{cases}$

Por tanto, $V_1(t) = \frac{\partial x}{\partial \lambda}(t, 0) = 1$

(ii) $\ddot{x} + x^3 = 0, \quad x(0) = 0, \quad \dot{x}(0) = \lambda$

Solo cambia el cálculo de $V(0)$ respecto al apartado anterior.

$V(0) = \frac{dz}{d\lambda}(0, 0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow \begin{cases} C_2 = 0 \\ C_1 = 1 \end{cases} \Rightarrow V_1(t) = \frac{\partial x}{\partial \lambda}(t, 0) = t$

$$(iii) \ddot{x} + \lambda x^3 = 0, \quad x(0) = \dot{x}(0) = 0$$

$$\text{Sea } x_1 = x, x_2 = \dot{x}, z = (x_1, x_2), D = \mathbb{R} \times \mathbb{R}^2, \Lambda = \mathbb{R}$$

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = -\lambda x_1^3 \quad \Sigma: D \times \Lambda \rightarrow \mathbb{R} \quad / \quad \Sigma(t, z, \lambda) = \begin{pmatrix} x_2 \\ -\lambda x_1^3 \end{pmatrix}$$

$$\frac{\partial \Sigma}{\partial z}(t, z, \lambda) = \begin{pmatrix} 0 & 1 \\ -3\lambda x_1^2 & 0 \end{pmatrix}, \quad \frac{\partial \Sigma}{\partial \lambda}(t, z, \lambda) = \begin{pmatrix} 0 \\ -x_1^3 \end{pmatrix} \quad \text{const.} \Rightarrow 2 \text{ dig.}$$

$$\frac{\partial \Sigma}{\partial z}(t, z(t, 0), 0) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = A \quad \frac{\partial \Sigma}{\partial \lambda}(t, z(t, 0), 0) = \begin{pmatrix} 0 \\ -x_1^3(t, 0) \end{pmatrix}$$

$$\ddot{x} = 0 \Rightarrow x(t) = c_1 t + c_2, \quad \dot{x}(t) = c_1 \Rightarrow c_1 = c_2 = 0 \Rightarrow x(t, 0) = 0 \Rightarrow$$

$$\frac{\partial \Sigma}{\partial \lambda}(t, z(t, 0), 0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = B$$

$$\text{Sea } V(t) = \begin{pmatrix} \frac{\partial x}{\partial \lambda}(t, 0) \\ \frac{\partial \dot{x}}{\partial \lambda}(t, 0) \end{pmatrix}$$

Por la diferenciabilidad respecto a parámetros,

$$\dot{V}(t) = A V + B \Rightarrow \begin{pmatrix} \dot{V}_1 \\ \dot{V}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} \Rightarrow \begin{cases} \dot{V}_1 = V_2 \\ \dot{V}_2 = 0 \end{cases} \Rightarrow$$

$$V_1(t) = c_1 t + c_2$$

$$V_2(t) = c_1$$

$$\frac{dz}{d\lambda}(t, 0) = \frac{\partial z}{\partial t}(t, 0) \cdot \frac{dt}{d\lambda}(0) \Big|_{\lambda=0} + \frac{\partial z}{\partial \lambda}(t, 0) + \frac{dz}{d\lambda} = V(t)$$

$$\frac{dz}{d\lambda}(0, 0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = V(0) \Rightarrow c_1 = c_2 = 0 \Rightarrow \frac{\partial x}{\partial \lambda}(t, 0) = V_1(t) = 0$$

$$(w) \quad \ddot{x} + x^3 = \lambda, \quad x(0) = \dot{x}(0) = 0.$$

$$\text{Sea } x_1 = x, x_2 = \dot{x}, z = (x_1, x_2), D = \mathbb{R} \times \mathbb{R}^2, \Lambda = \mathbb{R}$$

$$x_1 = \dot{x} = x_2 \quad \Sigma: D \times \Lambda \rightarrow \mathbb{R} \quad / \quad \Sigma(t, z, \lambda) = \left(\begin{matrix} x_2 \\ \lambda - x_1^3 \end{matrix} \right)$$

$$\frac{\partial \Sigma}{\partial z}(t, z, \lambda) = \begin{pmatrix} 0 & 1 \\ -3x_1^2 & 0 \end{pmatrix}, \quad \frac{\partial \Sigma}{\partial \lambda}(t, z, \lambda) = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{cont.} \Rightarrow z \text{ dif.}$$

$$\frac{\partial \Sigma}{\partial z}(t, z(t, 0), 0) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = A \quad \frac{\partial \Sigma}{\partial \lambda}(t, z(t, 0), 0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = B, \text{ pues}$$

$$x(t, 0) = 0 \quad \text{es sol. de } \begin{cases} \ddot{x} + x^3 = 0 \\ x(0) = \dot{x}(0) = 0 \end{cases}$$

$$\text{Sea } V(t) = \begin{pmatrix} \frac{\partial x}{\partial \lambda}(t, 0) \\ \frac{\partial \dot{x}}{\partial \lambda}(t, 0) \end{pmatrix}$$

Por la diferenciabilidad respecto a parámetros,

$$\dot{V}(t) = A V + B \Rightarrow \begin{pmatrix} \dot{V}_1 \\ \dot{V}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow$$

$$\begin{cases} \dot{V}_1 = V_2 \\ \dot{V}_2 = 1 \end{cases} \Rightarrow \ddot{V}_1 = \dot{V}_2 = 1 \Rightarrow \dot{V}_1(t) = t + C_1 \Rightarrow V_1(t) = \frac{t^2}{2} + C_1 t + C_2$$

$$\frac{dz}{d\lambda}(t, 0) = \frac{\partial z}{\partial t}(t, 0) \cdot \frac{d}{d\lambda}(0) \Big|_{\lambda=0} + \frac{\partial z}{\partial \lambda}(t, 0) \cdot \frac{d\lambda}{d\lambda}(0) = V(t) \Rightarrow$$

$$V(0) = \frac{dz}{d\lambda}(0, 0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow V_1(t) = \frac{\partial x}{\partial \lambda}(t, 0) = \frac{t^2}{2}$$

(36) Sea $x(t; a, b)$ la solución del problema

$$\dot{x} + \sin(ax) = t, \quad x(0) = b.$$

Demuestra

- (i) $x(t; a, b)$ es única y está definida en $]-\infty, +\infty[$
- (ii) $(t; a, b) \in \mathbb{R}^3 \mapsto x(t; a, b) \in \mathbb{R}$ es diferenciable
- (iii) Calcula $\frac{\partial x}{\partial a}(t; 0, 1)$ y $\frac{\partial x}{\partial b}(t; 0, 1)$.

$$\text{Sea } D = \mathbb{R}^2, 1 = \mathbb{R}$$

$$\text{Sea } \Sigma: D \times 1 \rightarrow \mathbb{R} / \Sigma(t, x, a) = t - \sin(ax)$$

$\frac{\partial \Sigma}{\partial x}(t, x, a) = -a \cos(ax)$ continua y es ≠ 0 hay unicidad $\forall (t_0, x_0) \in D \Rightarrow$ es especial para (0,0).

$\Rightarrow$ Σ tiene crecimiento lineal $\forall a \in 1 \Rightarrow$

Además, $|\Sigma(t, x, a)| \leq |t| + |\sin(ax)| \leq |t| + |x| \Rightarrow$ la sol. está definida en $\mathbb{R}$.

Como $\exists \frac{\partial \Sigma}{\partial x}(t, x, a), \frac{\partial \Sigma}{\partial a}(t, x, a) = -x \cos(ax)$ y son cont. $\Rightarrow x = x(t, a, b)$ es dig. por 1° de diferenciability respecto a condiciones iniciales y parámetros.

1) Calculamos $\frac{\partial x}{\partial a}(t, 0, 1)$

$$\frac{\partial \Sigma}{\partial x}(t, x(t, a, b), a) = -a \cos(ax(t, a, b)) \Rightarrow \frac{\partial \Sigma}{\partial x}(t, x(t, 0, 1), 0) = 0$$

$$\frac{\partial \Sigma}{\partial a}(t, x(t, 0, 1), 0) = y(t), \text{ donde } \Sigma \text{ es la única sol. de}$$

$$\begin{cases} \dot{y}(t) = \frac{\partial \Sigma}{\partial x}(t, x(t, 0, 1), 0) + \frac{\partial \Sigma}{\partial a}(t, x(t, 0, 1), 0) = 0 \\ y(0) = 0 \end{cases} \Rightarrow \dot{y} = -x$$

$$\text{Calculamos } x(t, 0, 1): \begin{cases} \dot{x} = t \\ x(0) = 1 \end{cases} \Rightarrow x(t) = \frac{t^2}{2} + 1, \quad \begin{cases} \dot{x} = t \\ x(0) = 1 \end{cases} \Rightarrow x(t) = \frac{t^2}{2} + 1$$

$$\dot{y} = -\frac{t^2}{2} - 1 \Rightarrow \begin{cases} y(t) = -\frac{t^3}{6} - t + C_1 \\ y(0) = C_1 = 0 \end{cases} \Rightarrow y(t) = -\frac{t^3}{6} - t$$

$$\text{Por tanto, } \frac{\partial x}{\partial a}(t, 0, 1) = -\frac{t^3}{6} - t$$

2) Calculamos $\frac{\partial x}{\partial b}(t, 0, 1)$

$$\frac{\partial \Sigma}{\partial b}(t, 0, 1) = \dot{\Sigma}(t), \text{ donde } \Sigma(t) \text{ es la única sol. de}$$

$$\begin{cases} \dot{\Sigma}(t) = \frac{\partial \Sigma}{\partial x}(t, x(t, 0, 1), 0) \Sigma \\ \Sigma(0) = \Sigma_a = 1 \end{cases} \Rightarrow \dot{\Sigma}(t) = 0 \Rightarrow \Sigma(t) = C_1$$

$$\Sigma(0) = \Sigma_a = 1 \Rightarrow \Sigma(t) = 1$$

$$\text{Por tanto, } \frac{\partial x}{\partial b}(t, 0, 1) = 1$$